

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

Roll No. :34	Name: Mehek Sawant
Class : Sy-It	Batch : 02
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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
arr=[11,12,13,14]
```

```
def enqueue(data):
```

```
    ele=int(input("Enter the Element= "))
```

```
    arr.append(ele)
```

```
def dequeue(data):
```

```
    arr.pop(data)
```

```
def show():
```

```
    print(arr)
```

```
num=0
```

```
while(num<4):
```

```
    print("1.Input 2.Delete 3.Show ")
```

```
    num= int(input("Select Choice: "))
```

```
    if(num==1):
```

```
        enqueue(5)
```

```
    elif(num==2):
```

```
        dequeue()
```

```
    elif(num==3):
```

```
        show()
```

```
    else:
```

```
print("Invalid command")
```

Output:

```
.PY
1.Input 2.Delete 3.Show
Select Choice: 1
Enter the Element= 12
1.Input 2.Delete 3.Show
Select Choice: 3
[11, 12, 13, 14, 12]
1.Input 2.Delete 3.Show
Select Choice: 3
[11, 12, 13, 14, 12]
```

b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```
import time
que=[("Cus1", 2), ("Cus2", 7), ("Cus3", 1)]

while que:
    job, timer= que.pop()
    print(f"----Printing---: {job} {timer}")
    time.sleep(timer)
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/De
.PY
----Printing---: Cus3 1
----Printing---: Cus2 7
----Printing---: Cus1 2
>>>
```

